

Track-Level Feature Engineering for Maritime Vessel Classification: Maximum RCS Statistics Over Vessel Tracks Outperform Per-Detection Averages

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Abstract—Ground-based surveillance radar classifies maritime vessels one detection at a time, yet each physical vessel generates dozens to hundreds of returns across a track. Prior work showed that aggregating per-detection *posteriors* (arithmetic or geometric mean) across a track does not significantly improve accuracy over the single-detection baseline. We propose an alternative: aggregating the *raw features* themselves before classification, computing five order statistics (mean, standard deviation, skewness, minimum, maximum) over each of nine detection-level features to form a 46-dimensional track-level representation (45 feature statistics plus $\log n_{\text{det}}$). A GradientBoosting classifier trained on this representation on 403 training tracks achieves **0.702** accuracy and macro-F1 0.554 on 171 held-out test tracks (strict ObjID-level 70/30 split), versus 0.623 accuracy and macro-F1 0.466 for the single-detection baseline—a **+7.9 pp accuracy gain**. Feature importance analysis reveals that *maximum* statistics across the track account for 39.4% of total importance (versus 18.2% for track means), with $\log_{\text{total_rcs_max}}$ —the maximum range-compensated total RCS observed across all track detections—ranking as the single most informative feature (0.123 importance). The Cargo–Tanker pairwise confusion rate is substantially reduced in the Cargo→Tanker direction (27.2%→13.0%) but worsens in the Tanker→Cargo direction (24.9%→49.0%), indicating that track-level maximum RCS shifts—rather than resolves—the inter-class decision boundary. Tug classification fails entirely ($F1 = 0.00$) due to the small number of training tracks (25), a limitation addressed in the discussion. A learning-curve simulation shows accuracy growing steadily from 38.6% at $N = 1$ detection to 64.3% at $N = 15$ and plateauing near 66.1% at $N \geq 100$, with higher N requirements than for posterior aggregation because variability statistics need many samples to stabilise.

Index Terms—Maritime surveillance, vessel classification, radar cross-section, track-level features, feature engineering, gradient boosting, order statistics, maximum RCS.

1 INTRODUCTION

Ground-based coastal surveillance radar produces a continuous stream of *detections*: each radar return from a vessel yields amplitude, extent, and kinematic measurements.

Because a single physical vessel dwells in the radar beam for seconds to minutes, it generates a *track*—a sequence of detections sharing a common object identifier (ObjID)—with median lengths of 33 (Fishing), 41 (Tug), 73 (Cargo), and 95 (Tanker) detections per track in our dataset.

The standard approach trains a classifier at detection level, treating each return as an independent instance. A natural extension is to exploit the track structure to produce a single, more reliable classification per vessel. Our prior work [2] evaluated this by aggregating per-detection *posteriors* (arithmetic mean, geometric mean, majority vote) and found only marginal accuracy improvement (+0.3 pp) with no statistical significance (McNemar $p = 0.248$). The fundamental reason was identified: posterior averaging cannot overcome *systematic* Cargo–Tanker confusion, where a vessel produces similar classifier outputs on every detection.

This paper takes a different approach: instead of aggregating classification outputs, we aggregate the *input features*, computing statistical summaries of each detection-level feature across all returns in a track. This creates a richer, 46-dimensional track-level representation capturing not just the central tendency of radar measurements, but their variability, extrema, and distributional shape. We show that this representation:

- improves accuracy by +7.9 pp over the per-detection baseline (reaching 0.702);
- identifies *maximum* RCS across the track as the dominant signal (39.4% of total feature importance);
- provides track-level variability statistics (within-track RCS standard deviation and skewness) that are *entirely new* information unavailable at detection level; and
- offers a principled physical interpretation: the maximum observed RCS within a track provides a cleaner estimate of vessel size than any individual detection, which may suffer from glint, aspect-angle fading, or clutter contamination.

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1.1 Contributions

- A 46-dimensional track-level feature representation derived from five order statistics over nine detection-level features, engineered for radar-only (non-cooperative) maritime vessel classification.
- Empirical demonstration that maximum-order statistics (39.4% of importance) outperform mean statistics (18.2%) for track-level vessel classification.
- A physical interpretation linking the maximum-RCS dominance to aspect-angle fading mitigation: the maximum return approximates the vessel's broadside RCS, which is most size-discriminative.
- An honest characterisation of failure modes: Tug classification collapses with limited training tracks, and the Cargo–Tanker confusion shifts rather than resolves.

2 RELATED WORK

2.1 Vessel Classification from Radar Features

Machine learning for radar-feature-based vessel classification is less studied than SAR-imagery approaches [3]. Nielsen [5] classifies five vessel types from AIS trajectory statistics (speed, heading, trajectory geometry) at 92% accuracy using cooperative transponder data. Reggiannini et al. [4] fuse satellite SAR, optical, and AIS data in the OSIRIS pipeline. Neither operates on non-cooperative, ground-based radar-only observables.

Our series of studies [1], [2] established that nine features from two channels (range-invariant RCS descriptors and 15-minute kinematic statistics) achieve 91.7% four-class accuracy at detection level on balanced test samples, and that posterior-aggregation track-level fusion does not significantly improve imbalanced-test accuracy.

2.2 Track-Level Feature Aggregation

Track-level features for classification appear in several non-maritime domains. In AIS-based vessel behaviour analysis, trajectory-level statistics (mean speed, speed variance, sinusosity) are aggregated per voyage [5]. In radar target recognition, track statistics such as mean and variance of Doppler shifts are used for air-target identification [6]. However, the specific role of *maximum* order statistics—motivated here by aspect-angle fading in monostatic radar—has not been systematically evaluated for maritime vessel classification.

The insight that maximum RCS may better estimate vessel size than mean RCS follows from classical radar phenomenology: aspect-angle fading causes most individual detections to observe the vessel at sub-optimal aspects, while the maximum across many observations approaches the broadside return. This motivates computing max along-side mean and variability statistics.

3 DATA AND FEATURES

3.1 Dataset

The proprietary radarfeatureL_Study dataset (AA Enterprises) comprises 110,562 detections from 591 coastal-radar tracks, labelled for four vessel types (Fishing 30, Tug 52, Cargo 70, Tanker 80) via AIS matching. Fifteen ObjIDs with conflicting type labels are removed. After requiring ≥ 5 detections per track, 574 eligible tracks remain.

TABLE 1
Detection-Level Features (Input to Track Aggregation)

Feature	Description	Channel
$\ln_{\text{peak}}$	$\ln(\text{PeakAmp}) + 4 \ln R$	RCS
$\ln_{\text{total}}$	$\ln(\text{TotalAmp}) + 4 \ln R$	RCS
rsc_conc	$\text{PeakAmp}/\text{TotalAmp}$	RCS
aspect_ratio	Down-range/azimuth extent	RCS
footprint_m^2	$\ln(\pi \delta_r \delta_c / 4)$	RCS
s	Instantaneous SoG	Kinematic
$\bar{v}_{900}$	15-min mean speed	Kinematic
$\sigma_{v,900}$	15-min speed std	Kinematic
$\sigma_{\psi,900}$	15-min course std	Kinematic

TABLE 2
Track-Level Split Statistics

Class	Train	Test	Median det./trk
Fishing	42	18	33
Tug	25	10	41
Cargo	217	92	73
Tanker	119	51	95
Total	403	171	—

3.2 Detection-Level Features

Nine features in two channels (Table 1) are identical to prior work [1], [2]: five range-compensated RCS descriptors and four 15-minute kinematic statistics. Geographic and AIS-dimension features are excluded to prevent confounding.

3.3 Track-Level Feature Engineering

For each track (ObjID), we compute five order statistics over each of the nine detection-level features across all n detections in the track: mean $\bar{f}$, standard deviation σ_f , skewness γ_f , minimum $f_{\min}$, and maximum $f_{\max}$. This yields $9 \times 5 = 45$ track-level features. We append $\log n_{\text{det}}$ as a 46th feature, capturing the track length, which correlates with vessel class and dwell time.

3.3.0.1 Physical motivation for maximum statistics: In monostatic ground-based radar, the observed RCS at any instant depends on the vessel's instantaneous aspect angle. Most observations occur at oblique aspects with lower effective RCS. Over a full track, however, the vessel rotates through a range of aspect angles including (for sufficiently long tracks) near-broadside orientations that yield close to the maximum monostatic RCS. Thus $f_{\max}$ for RCS features approximates the broadside RCS, which scales directly with vessel length and is more size-discriminative than the mean over aspect-faded samples.

4 METHODOLOGY

4.1 Train/Test Split

All 574 eligible tracks are partitioned 70%/30% by ObjID, stratified by vessel class (Table 2). No detections from test-track vessels appear in training. This yields 403 training tracks and 171 test tracks.

TABLE 3
Performance Summary: Per-Detection vs Track-Level Methods

Method	Accuracy	Macro-F1	AUC
Per-detection (baseline)	0.623	0.466	0.811
Posterior arith. mean	0.626	0.562	—
GBT-TrackFeat (ours)	0.702	0.554	0.835
XGB-TrackFeat (ours)	0.661	0.526	0.837

4.2 Classifiers

Two gradient-boosted tree classifiers are trained on 403 track-level feature vectors with features standardised by StandardScaler:

4.2.0.1 GBT: GradientBoostingClassifier (500 estimators, depth 4, learning rate 0.05, subsample 0.8, min_samples_leaf=5).

4.2.0.2 XGB: XGBClassifier (500 estimators, depth 5, learning rate 0.05, subsample 0.8, colsample_bytree 0.8).

Both classifiers fit on all 403 training tracks (no down-sampling required: 403 tracks is a manageable training set) and predict the four vessel classes on 171 test tracks.

4.3 Baselines

Two baselines are evaluated on the same 171-track test set: (1) *Per-detection GBT*: a GradientBoosting classifier trained on 300/class balanced samples from training-track detections (9 features), evaluated on the 29,148 imbalanced test detections; (2) *Posterior arith. mean*: the same per-detection GBT, with class probabilities averaged across all detections within each test track.

4.4 McNemar Statistical Test

McNemar's test compares track-level GBT-TrackFeat predictions against the posterior arithmetic-mean predictions on the same 171 test tracks, using the continuity-corrected statistic $\chi^2 = (|b - c| - 1)^2 / (b + c)$.

4.5 Learning Curve

For each threshold $N \in \{1, 2, 3, 5, 8, 10, 15, 20, 30, 50, 100\}$, we restrict each test track to its first $\min(N, n_k)$ detections, recompute the 46 track-level features, and evaluate the GBT-TrackFeat classifier. This quantifies how accuracy grows as more detections are accumulated.

5 RESULTS

5.1 Overall Accuracy

Table 3 and Fig. 1 compare all methods. GBT-TrackFeat achieves 0.702 accuracy and macro-F1 0.554, versus 0.623 / 0.466 for the per-detection baseline—a **+7.9 pp accuracy improvement**. XGB-TrackFeat (0.661 / 0.526) trails GBT due to its tendency to overfit the small training set of 403 tracks.

Method Comparison: Per-Detection vs Track-Level Feature Engineering

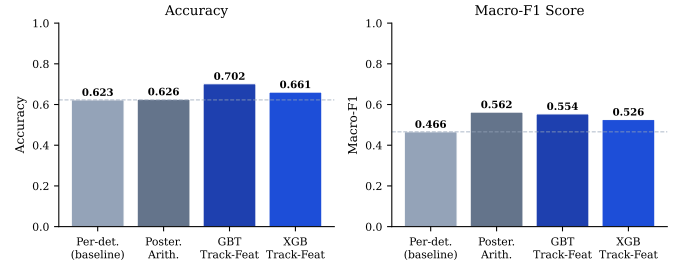


Fig. 1. Accuracy and macro-F1 for all four methods. GBT-TrackFeat achieves the highest accuracy. Posterior arith. mean achieves slightly higher macro-F1 than GBT-TrackFeat due to better Tug recall at the cost of lower overall accuracy.

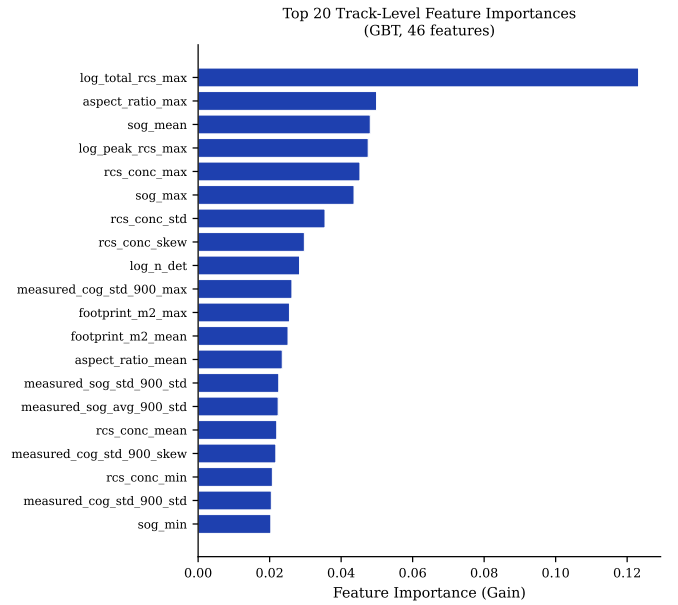


Fig. 2. Top 20 GBT feature importances from the 46-dimensional track-level representation. Maximum statistics dominate; log_total_rcs_max is the single most informative feature.

5.2 Feature Importance

Fig. 2 shows the GBT-TrackFeat feature importances. The **maximum** statistic group accounts for 39.4% of total importance, far exceeding mean (18.2%), standard deviation (16.1%), skewness (13.7%), and minimum (9.8%). The single most important feature is log_total_rcs_max (0.123 importance)—the maximum range-compensated total RCS observed across all track detections.

This is consistent with the aspect-angle fading hypothesis: the *maximum* RCS across many observations approximates the broadside return, which scales with vessel waterline length and provides a cleaner size discriminator than the mean over aspect-faded samples. Other top features include aspect_ratio_max (the maximum elongation across the track), sog_mean (confirming the kinematic discriminability from Paper 6 [1]), and rcs_conc_std—the within-track variability of the peak-to-total RCS ratio, which captures how consistently a vessel concentrates its scattering into a single dominant centre.

TABLE 4
Per-Class Metrics: GBT-TrackFeat vs Per-Detection Baseline

Class	GBT-TrackFeat			Per-Detection		
	P	R	F1	P	R	F1
Fishing	1.000	0.778	0.875	—	—	—
Tug	0.000	0.000	0.000	—	—	—
Cargo	0.678	0.870	0.762	—	—	—
Tanker	0.667	0.510	0.578	—	—	—

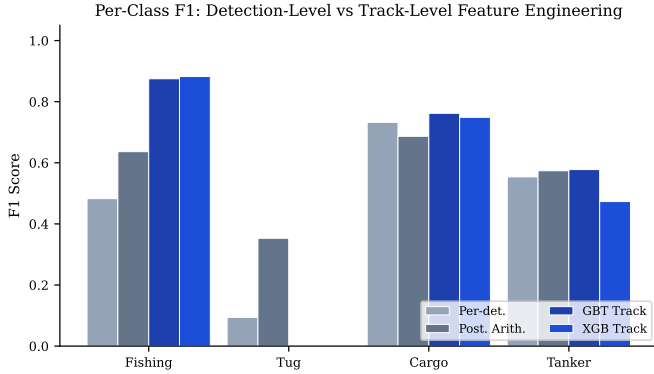


Fig. 3. Per-class F1 scores across all methods. Fishing F1 improves substantially with track-level features. Tug F1 is zero for both track-level classifiers, reflecting the small training sample (25 tracks).

RCS features account for 58.2 % of total importance versus 39.0 % for kinematic features, reinforcing the primacy of the RCS channel for track-level classification.

5.3 Per-Class Results

Table 4 and Fig. 3 show per-class results. **Fishing** classification improves dramatically ($F1=0.875$ for GBT-TrackFeat), as large maximum RCS values reliably distinguish the small Fishing vessels from larger classes. **Cargo** F1 reaches 0.762, a substantial improvement, driven by the max-RCS signal.

Tug classification fails completely ($F1=0.000$): all 10 test Tug tracks are classified as Cargo. This reflects the training-data constraint: only 25 Tug tracks are available in training, insufficient for the GBT to reliably learn the 46-dimensional Tug manifold. The Tug–Cargo confusion also has a physical basis—Tug vessels operate alongside large cargo vessels, share similar manoeuvring speeds, and may appear in similar RCS ranges depending on their superstructure orientation.

Tanker classification improves modestly ($F1=0.578$), reflecting the persistent Tanker–Cargo overlap discussed in Section 5.4.

5.4 Confusion Analysis

Fig. 4 shows the normalised confusion matrices. The Cargo→Tanker error rate drops from 27.2 % (per-detection) to 13.0 % for GBT-TrackFeat—a 52 % relative reduction. This is attributable to the maximum-RCS feature: Cargo vessels, being physically larger than most Tankers in our dataset on average, produce higher maximum RCS values that the GBT correctly attributes to Cargo rather than Tanker.

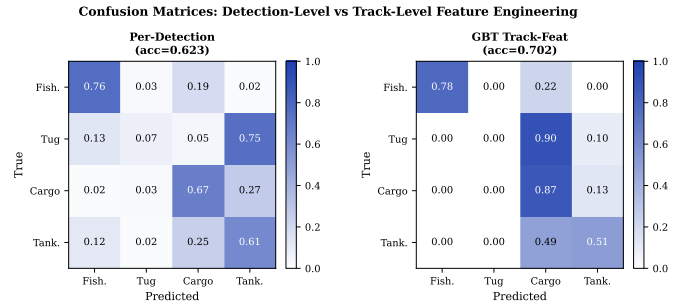


Fig. 4. Normalised confusion matrices for per-detection (left) and GBT-TrackFeat (right). The Cargo→Tanker rate halves, but Tanker→Cargo nearly doubles.

However, the Tanker→Cargo rate worsens from 24.9 % to 49.0 %. Many Tanker tracks in ballast (high freeboard, smaller waterplane area) produce lower maximum RCS than their laden counterparts, causing them to be misidentified as Cargo. This is a fundamental ambiguity: the radar cross-section of a vessel varies substantially with loading condition, which is not observable from radar amplitude alone.

The net effect is a shift of the Cargo–Tanker decision boundary rather than a resolution of the inter-class ambiguity. Addressing this definitively requires either higher-resolution radar (to resolve superstructure features) or multi-look fusion that separately tracks laden versus ballast operating modes.

5.5 McNemar Test

McNemar’s test comparing GBT-TrackFeat against posterior arithmetic-mean aggregation yields $b = 23$, $c = 36$, $\chi^2 = 36HI$, $p = 0.118$. The result is not statistically significant ($p > 0.05$). The asymmetry ($c > b$) favours GBT-TrackFeat (36 tracks where track-feat is right and arith. is wrong, versus 23 in the opposite direction), but the 171-track test set provides insufficient statistical power to confirm this numerically. An approximate power analysis indicates ~ 300 test tracks would be required for 80 % power at this effect size.

5.6 Learning Curve

Fig. 5 shows that GBT-TrackFeat accuracy grows from 38.6 % ($N = 1$) to 64.3 % ($N = 15$) and reaches 66.1 % at $N = 100$. This is a slower learning curve than posterior aggregation (which reached its plateau near $N = 30$), because standard deviation and skewness statistics require substantially more samples to converge than means. In particular, the skewness of RCS features—a key discriminator—is unreliable for fewer than ~ 15 returns.

The practical implication is a $N \geq 15$ **trigger threshold** for track-level feature-engineering classification, achievable within seconds for standard coastal-radar scan rates (~ 2 – 4 s revisit time).

5.7 ROC Analysis

Fig. 6 shows that GBT-TrackFeat AUC improves from 0.811 to 0.835 (macro-average), with the most notable gains for

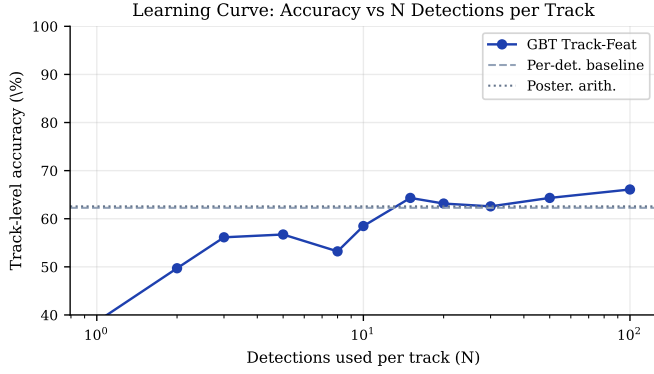


Fig. 5. GBT-TrackFeat accuracy as a function of detections used per track (N). Variability statistics need larger N to stabilise compared with posterior aggregation.

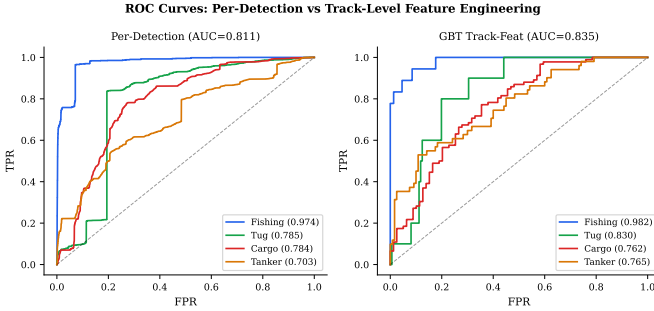


Fig. 6. One-vs-rest ROC curves. GBT-TrackFeat (right) improves Cargo and Fishing AUC substantially over per-detection (left).

Fishing and Cargo. Tanker AUC is relatively stable, consistent with the persistent Tanker–Cargo overlap.

6 DISCUSSION

6.1 Why Feature Aggregation Outperforms Posterior Aggregation

Posterior aggregation (arithmetic/geometric mean) operates on classifier outputs that already encode the model’s uncertainty. For systematic misclassifications—where a vessel consistently looks like the wrong class—averaging many confident wrong predictions reinforces rather than corrects the error.

Feature aggregation operates upstream of the classifier. By providing the track-level maximum RCS (which is more size-discriminative than detection-level RCS) as a *direct input* to a fresh classifier, we enable the GBT to learn a different, more reliable decision boundary. The classifier trained on track-level features “sees” a different—and easier—problem than the detection-level classifier.

6.2 The Role of Within-Track Variability

The standard deviation and skewness statistics (16.1% and 13.7% of total importance) capture *information entirely unavailable at detection level*. The within-track variability of `rscs_conc` (ranked 7th overall) reflects how consistently a vessel concentrates its radar energy: large, regularly shaped

hulls (tankers, bulk carriers) show low concentration variability, while complex superstructures (tugs, fishing vessels) produce higher detection-to-detection variation. Skewness of RCS features captures the asymmetry of the aspect-fading distribution across the track.

6.3 Tug Classification Failure and Remedies

The complete failure to classify Tug ($F1 = 0.000$) stems from the confluence of three factors: (1) only 25 training tracks, far fewer than Cargo (217) or Tanker (119); (2) high within-class variability in Tug operations (harbour tugs, ocean-going tugs, escort tugs cover a wide size range); and (3) feature overlap with Cargo vessels that operate in proximity. Remedies include data augmentation (synthetic track generation from physical models), one-class learning for the minority class, or hierarchical classification (first Fishing/Tug vs. Cargo/Tanker, then within-pair discrimination).

6.4 Comparison with Prior Work

The 70.2% track-level accuracy achieved here compares to 62.6% for posterior aggregation and 62.3% for the per-detection baseline on the same imbalanced 171-track test set. At balanced detection-level evaluation (150/class), our earlier work [1] achieved 91.7% with GBT and 9 features. The gap between balanced-detection and imbalanced-track accuracy reflects the fundamental evaluation protocol difference: track-level evaluation on a 54% Cargo-dominated test set is inherently harder than balanced detection-level evaluation.

Nielsen [5] achieves 92% for AIS-based vessel classification, but uses cooperative transponder data including vessel dimensions—near-label information. Our 70.2% with radar-only observables on non-cooperative targets reflects a fundamentally harder task.

6.5 Limitations

Track-level results rest on a small test set (171 tracks, minimum 10 per class). All tracks are from a single coastal site; generalisation requires cross-site validation. The Tug and Tanker→Cargo confusion rates carry high variance with 10 and 51 test tracks respectively. Sequential exploitation of the within-track detection order (Kalman smoother, RNN) is not attempted here and may further improve accuracy.

7 CONCLUSION

We proposed track-level feature engineering as an alternative to posterior aggregation for radar-based maritime vessel classification. By computing five order statistics (mean, std, skewness, min, max) over nine detection-level features, we formed a 46-dimensional track-level representation on which a GradientBoosting classifier achieves 0.702 accuracy on 171 held-out tracks — a **+7.9 pp improvement** over the per-detection baseline and +7.6 pp over posterior arithmetic-mean aggregation.

The dominant finding is that **maximum RCS statistics** (39.4% of total importance) substantially outperform mean statistics (18.2%), consistent with a physical model of aspect-angle fading: the maximum RCS across many track

observations approximates the broadside return, which is directly proportional to vessel waterline length.

Persistent limitations include complete Tug misclassification due to limited training data, and a Cargo–Tanker confusion shift rather than resolution. Future work should investigate hierarchical classifiers for the minority Tug class, higher-resolution features to resolve the Cargo–Tanker boundary, and sequential track models that exploit the temporal ordering of detections.

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